

# Agentic AI for Mechanism-Grounded Drug Repurposing

## 1. Overview

This project proposes an **agentic AI research assistant** designed to support *early-stage drug repurposing hypothesis exploration*. Unlike traditional systems that rely on similarity matching or large-scale literature summarization, this system prioritizes **biological and genetic plausibility first**, and then verifies hypotheses through evidence and cross-checks.

The system behaves like a **junior biomedical researcher**: it reasons over mechanisms of action, validates them across trusted biological sources, checks disease relevance, explores supporting and contradicting evidence, and produces an explainable, uncertainty-aware hypothesis report. The system does **not** claim therapeutic effectiveness; it supports scientific decision-making.

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## 2. Problem Statement

Drug repurposing is attractive due to reduced cost and faster timelines, but computational approaches often suffer from: - Over-reliance on similarity scores or black-box predictions - Limited biological interpretability - Poor handling of contradictory or negative evidence - Lack of explicit reasoning about mechanisms and uncertainty

This project addresses these limitations by building an **agentic, mechanism-driven, and evidence-traceable system** that evaluates whether a drug–disease pair is *biologically plausible* and worth further investigation.

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## 3. System Philosophy

**Core Principle:** > *Biology first, evidence second, conclusions last.*

Key design principles: - Mechanism-of-Action (MoA) is the entry gate - Cross-verification across multiple biological sources is mandatory - Literature and trials are used to strengthen or weaken hypotheses, not to invent them - Contradictions and uncertainty are first-class outputs - The system supports reasoning, not prediction

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## 4. User Input

The system takes a **minimal, hypothesis-driven input**:

Drug name

Disease name

Example:

Drug: Metformin

Disease: Alzheimer's disease

This input represents a single scientific hypothesis: > *Is it biologically plausible that this drug could impact this disease?*

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## 5. High-Level Architecture

The system consists of: - A **Master Orchestrator** - Two parallel reasoning tracks - Specialized agents responsible for distinct scientific tasks

### Reasoning Tracks

1. **Mechanism-First Track (Primary, high-confidence)**
2. **Phenotype / Evidence-First Track (Exploratory, low-confidence)**

The orchestrator coordinates execution and merges insights from both tracks.

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## 6. Agent Definitions & Responsibilities

### 6.1 Master Orchestrator

- Controls execution order
  - Maintains hypothesis state
  - Decides when to iterate, downgrade confidence, or stop exploration
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### 6.2 MoA & Target Enumeration Agent

**Purpose:** Establish biological grounding.

**Functions:** - Identify all known drug targets (primary and off-target) - Map targets to pathways and biological processes - Include tissue-specific expression where available

**Outputs:** - Candidate MoA chains with source attribution - Initial confidence indicators

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### 6.3 MoA Cross-Verification & Conflict Mapping Agent

**Purpose:** Validate mechanisms and detect inconsistencies.

**Functions:** - Cross-check MoA across multiple databases - Apply source hierarchy (experimental > curated > inferred) - Explicitly map conflicting or uncertain claims

**Outputs:** - Verified MoA list (strong / medium / weak) - Conflict and uncertainty annotations

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## 6.4 Disease–Mechanism Relevance Agent

**Purpose:** Assess whether mechanisms matter for the disease.

**Functions:** - Evaluate gene–disease and pathway–disease associations - Assess directionality of modulation (beneficial vs harmful) - Consider disease stage relevance

**Outputs:** - Disease relevance score - Directional confidence - Biological alignment notes

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## 6.5 Phenotype & Lightweight Literature Scan Agent

**Purpose:** Capture unexpected or serendipitous signals.

**Functions:** - Scan literature for indirect effects, failed trials, or off-label mentions - Identify contradictory or negative findings - Flag unexpected clinical observations

**Constraints:** - Cannot create mechanisms - Used only to trigger re-exploration or caution

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## 6.6 Clinical, PK & Safety Context Agent

**Purpose:** Ground hypotheses in real-world feasibility.

**Functions:** - Review trial outcomes (positive, failed, terminated) - Identify safety risks and contraindications - Evaluate PK/PD feasibility (target tissue accessibility)

**Outputs:** - Safety flags - Clinical feasibility notes - Known failure modes

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## 6.7 Prior Knowledge & Memory Agent

**Purpose:** Avoid rediscovering known failures.

**Functions:** - Track recurring mechanisms and outcomes - Detect historically unsuccessful patterns - Provide context for “why this hasn’t worked before”

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## 6.8 Hypothesis Synthesis & Uncertainty Modeling Agent

**Purpose:** Final reasoning and reporting.

**Functions:** - Integrate outputs from all agents - Generate structured rationale chains - Quantify uncertainty and confidence ranges - Propose competing hypotheses

**Outputs:** - Viability profile (not binary verdict) - Evidence-traceable hypothesis report - Suggested next experimental steps

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## 7. Workflow Summary

1. User submits Drug + Disease
  2. Orchestrator initializes hypothesis
  3. Mechanism-first track runs (MoA → verification → disease relevance)
  4. Phenotype track runs in parallel (lightweight evidence scan)
  5. Clinical & safety checks applied
  6. Hypothesis synthesis with uncertainty modeling
  7. Final structured output generated
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## 8. Output Structure

Each hypothesis report includes: - Mechanistic rationale chain - Evidence sources and citations - Conflicting findings and uncertainty - Safety and feasibility notes - Plausibility confidence range - Knowledge gaps and next experiments

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## 9. Intended Use & Limitations

**Intended Use:** - Early-stage hypothesis exploration - Research decision support - Academic and educational settings

**Limitations:** - Not a clinical decision tool - Dependent on open data completeness - Biological plausibility does not imply efficacy

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## 10. Conclusion

This system represents a shift from result-driven computational repurposing toward **agentic, mechanism-grounded scientific reasoning**. By prioritizing biological plausibility, cross-verification, and uncertainty awareness, it provides a transparent and responsible framework for exploring drug repurposing hypotheses.